# Separation Kernel Hypervisor

## Secure, Isolated and Predictable Computing

Bilvatech provides Separation Kernel Hypervisor solutions that enable multiple applications, operating systems, and workloads to run on a shared computing platform while maintaining clearly defined boundaries between them.

Our technology establishes controlled access to **CPU, memory, I/O, peripherals, and communication resources**, allowing different software environments to operate according to their specific requirements. This provides a foundation for building computing platforms with greater **security, predictability, fault containment, and resource efficiency**.

## Core Capabilities

**Strong System Isolation**
Bilvatech creates independent execution environments for applications and operating systems, helping contain software faults and prevent unauthorized access or interference between workloads.

**Resource Management**
We provide controlled allocation of processor cores, memory, peripherals, interrupts, and I/O resources, ensuring that each workload operates within its defined resource boundaries.

**Mixed-Criticality Workloads**
Our solutions enable applications with different safety, security, performance, and timing requirements to operate on the same hardware platform while maintaining appropriate levels of separation.

**Deterministic Execution**
Bilvatech supports predictable resource allocation for real-time and time-sensitive applications, helping maintain consistent system behavior where timing and performance are critical.

**Controlled Communication**
We establish defined communication paths between selected software environments, enabling required data exchange while preserving isolation between individual domains.

**Platform Consolidation**
Bilvatech enables multiple computing functions to be consolidated onto a common hardware platform, helping reduce hardware complexity, system footprint, and resource requirements.

## Built for Mission-Critical Computing

Our Separation Kernel Hypervisor solutions are designed for environments where **security, reliability, predictable performance, and controlled resource access** are essential. They support architectures that bring together real-time applications, operating systems, communications, monitoring, control, and data-processing workloads within a structured computing environment.

The technology is suited for **aerospace, defense, automotive, industrial automation, robotics, transportation, and other advanced embedded systems** where dependable and isolated execution is essential.

## From Architecture to Deployment

Bilvatech works across the solution lifecycle, from **system architecture and hardware mapping to software integration, configuration, verification, and deployment**. We help organizations develop computing platforms that can accommodate evolving software requirements while maintaining the isolation, control, and performance expected from mission-critical systems.

**Multiple workloads. One platform. Controlled by design.**